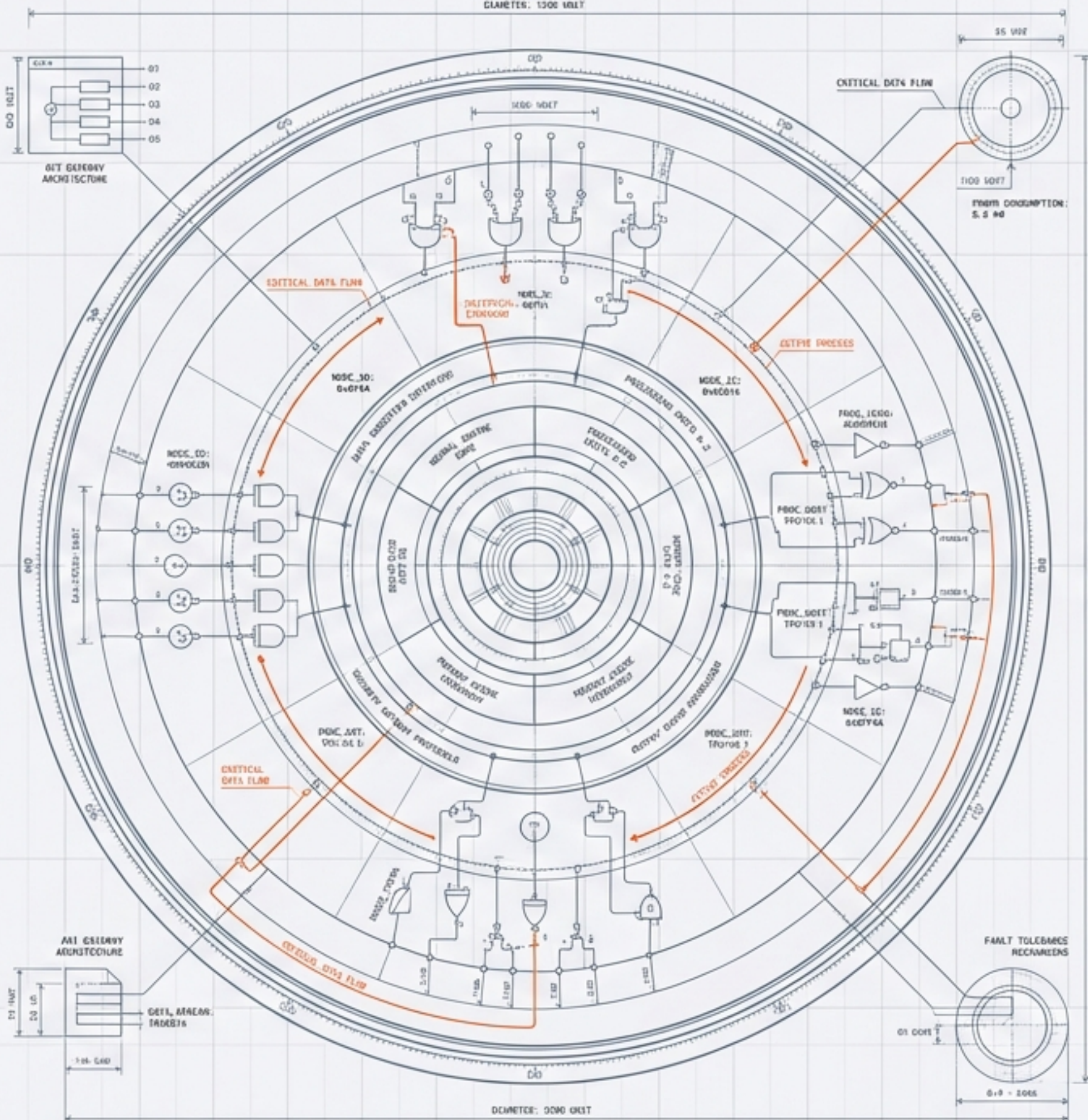


Intelligent AI-Powered Orchestration Platform

Phases F-J Complete

DOC REF: Architecture Specification & Implementation Report



Rigorous Governance & Autonomous Orchestration

CORTEX is a governance-aware development platform. It routes user intent through intelligent classification, applies immutable safety rules, and executes operations with full audit trails.



Governance-First



Resilience-First



Knowledge-Driven

BUSINESS NEED:

Consistent Code Quality

CORTEX SOLUTION: 29 Immutable TIER
0 Governance Rules

BUSINESS NEED:

Predictable Automation

CORTEX SOLUTION: 4-Stage Orchestration
Pipeline (ConversationProtocol)

BUSINESS NEED:

Audit Compliance

CORTEX SOLUTION: Hash-Chain
Verified Audit Logs

BUSINESS NEED:

Knowledge Reuse

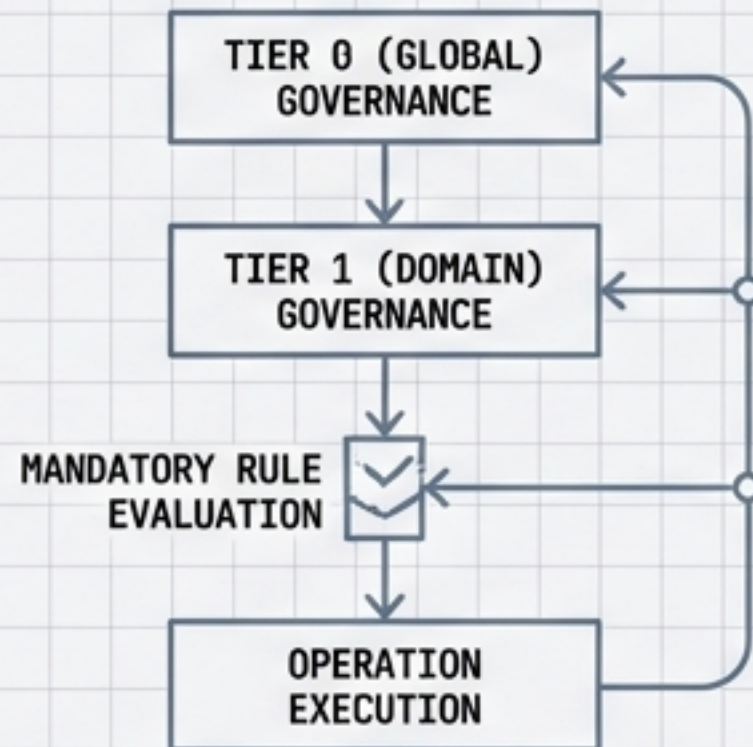
CORTEX SOLUTION: Domain Brain with
Conflict Resolution

Design Philosophy

“Safety is not a feature—it’s the foundation.”

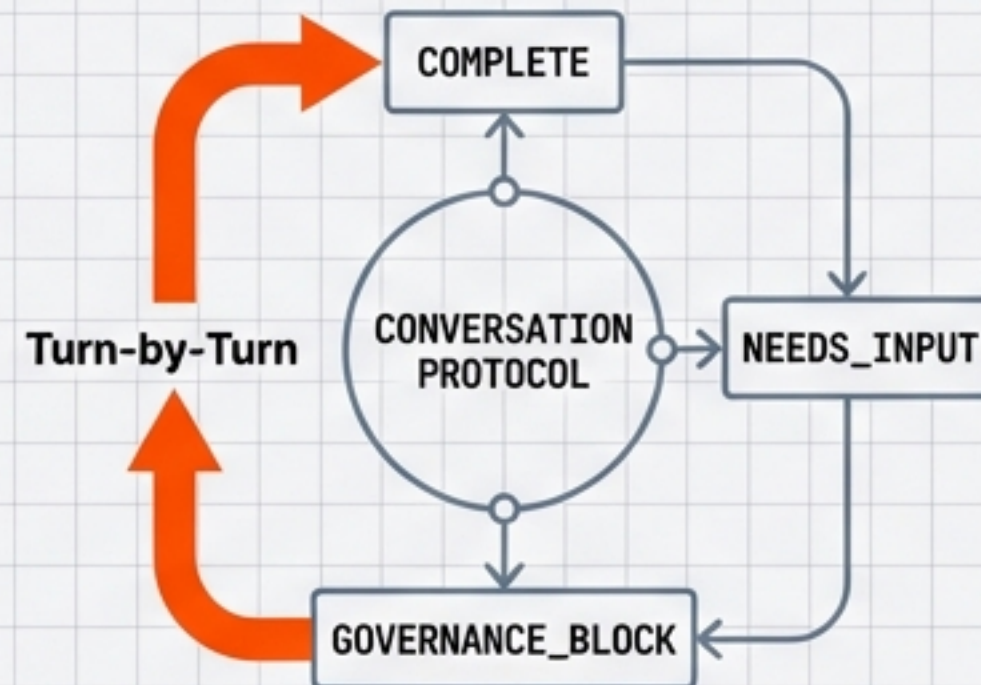
Governance-First Architecture

- Rule evaluation is mandatory.
- No operation bypasses governance.
- Tier 0 (Global) > Tier 1 (Domain).



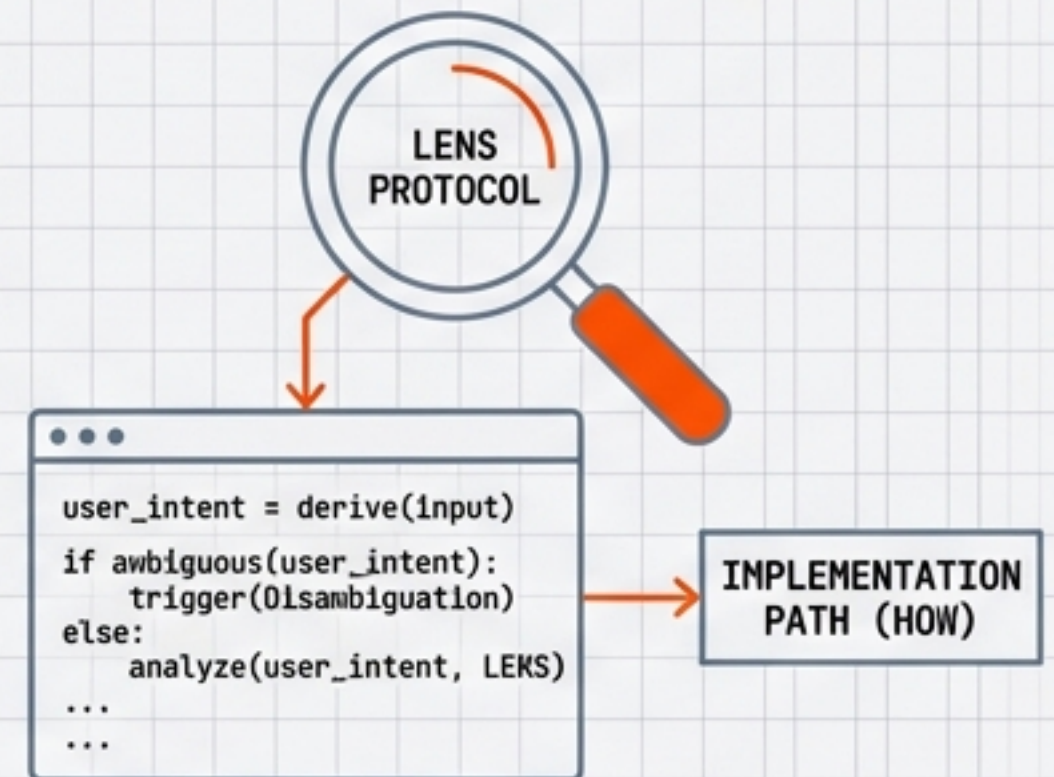
Orchestration as Conversation

- Operations are dialogues, not commands.
- Protocol: ConversationProtocol
- Decisions: COMPLETE, NEEDS_INPUT, GOVERNANCE_BLOCK

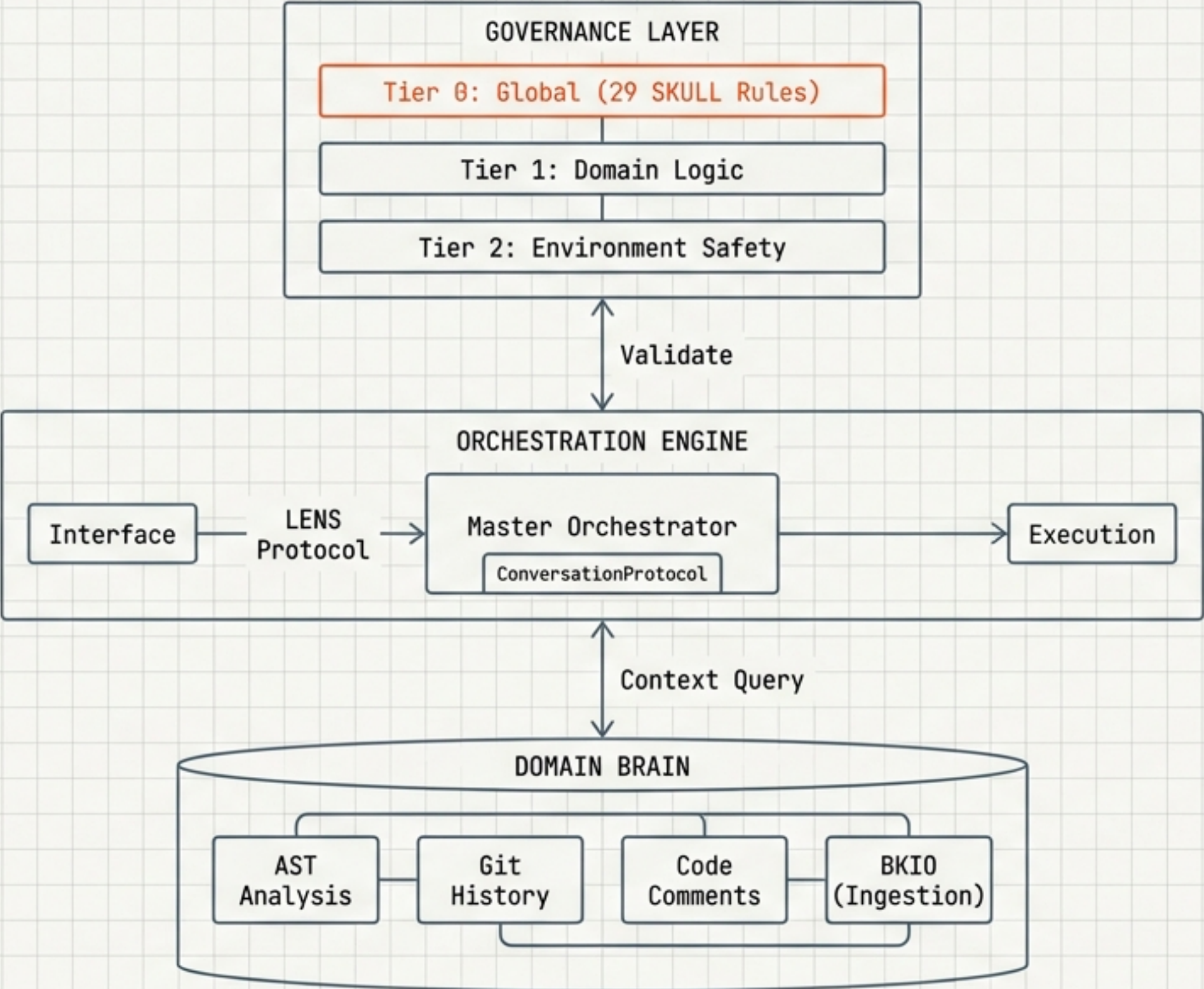


Intent Before Implementation

- Understand 'what' before deciding 'how'.
- Ambiguity triggers Disambiguation.
- LENS Protocol analysis.



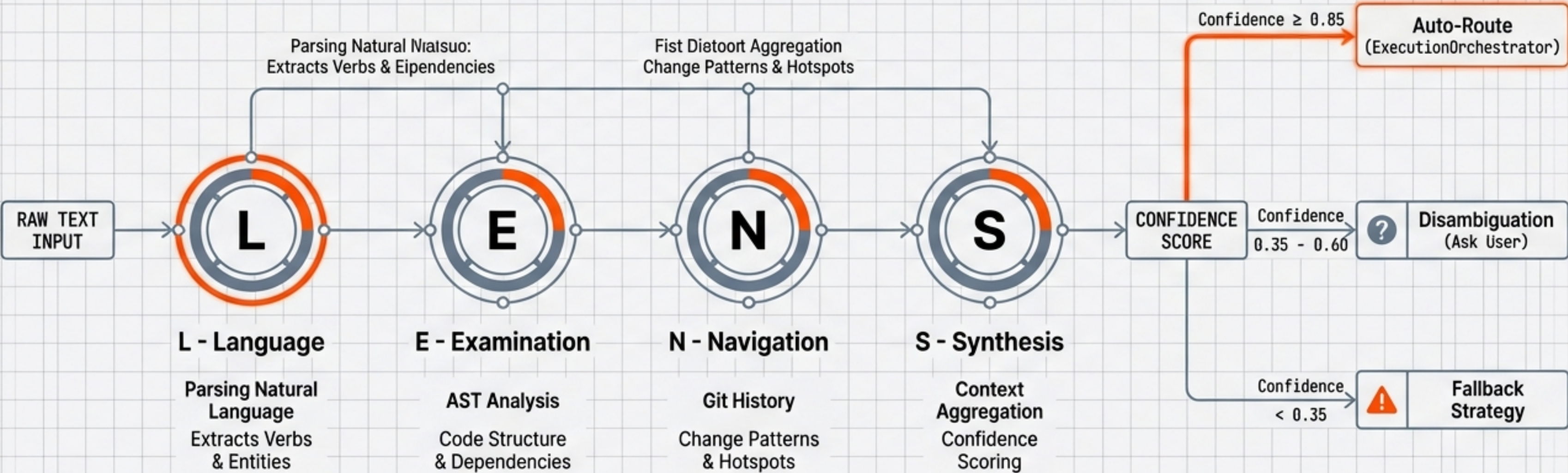
System Architecture



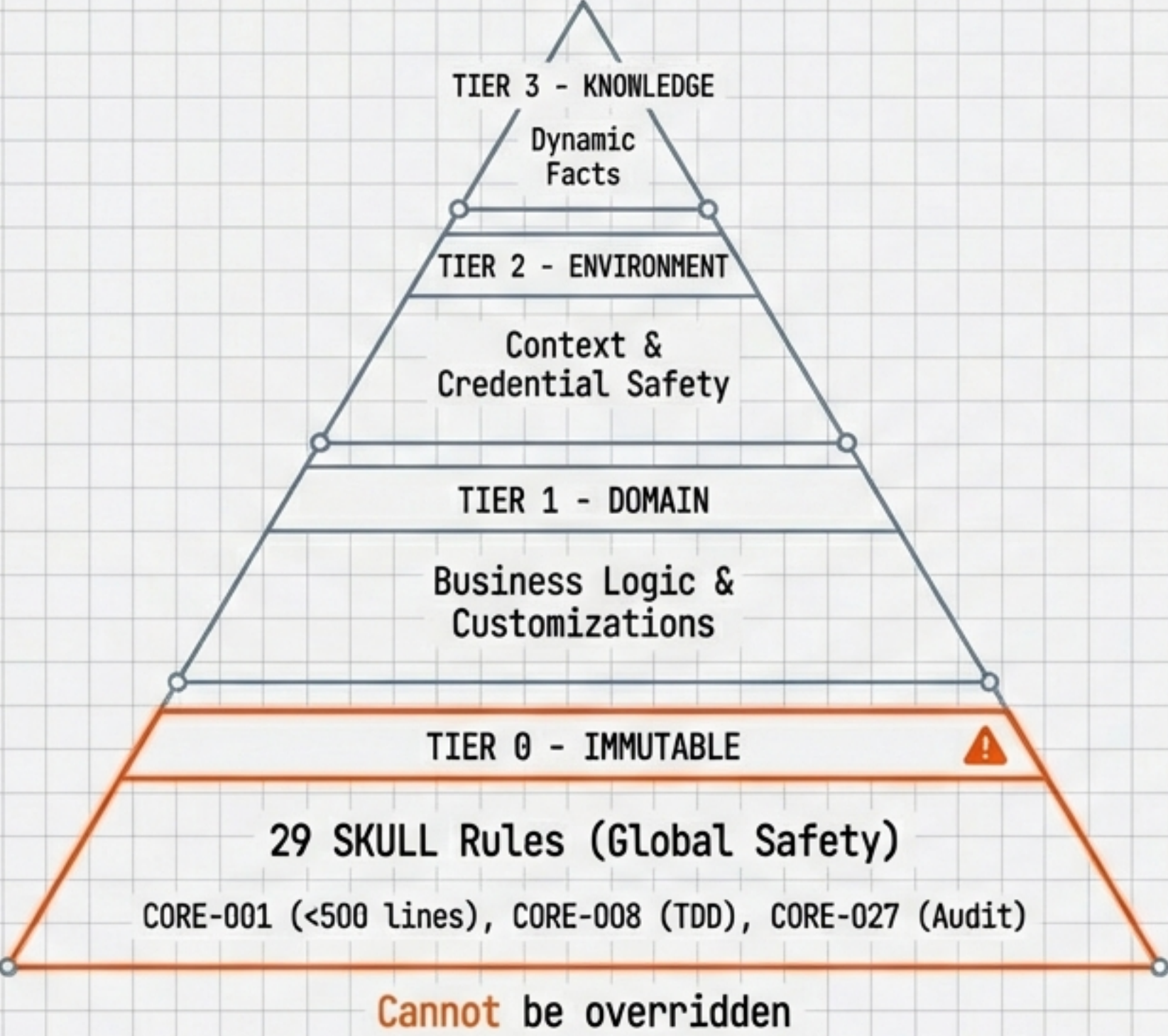
LENS Protocol: Intent Intelligence

Router Functional Status: 100% (128/128 Tests Passing)

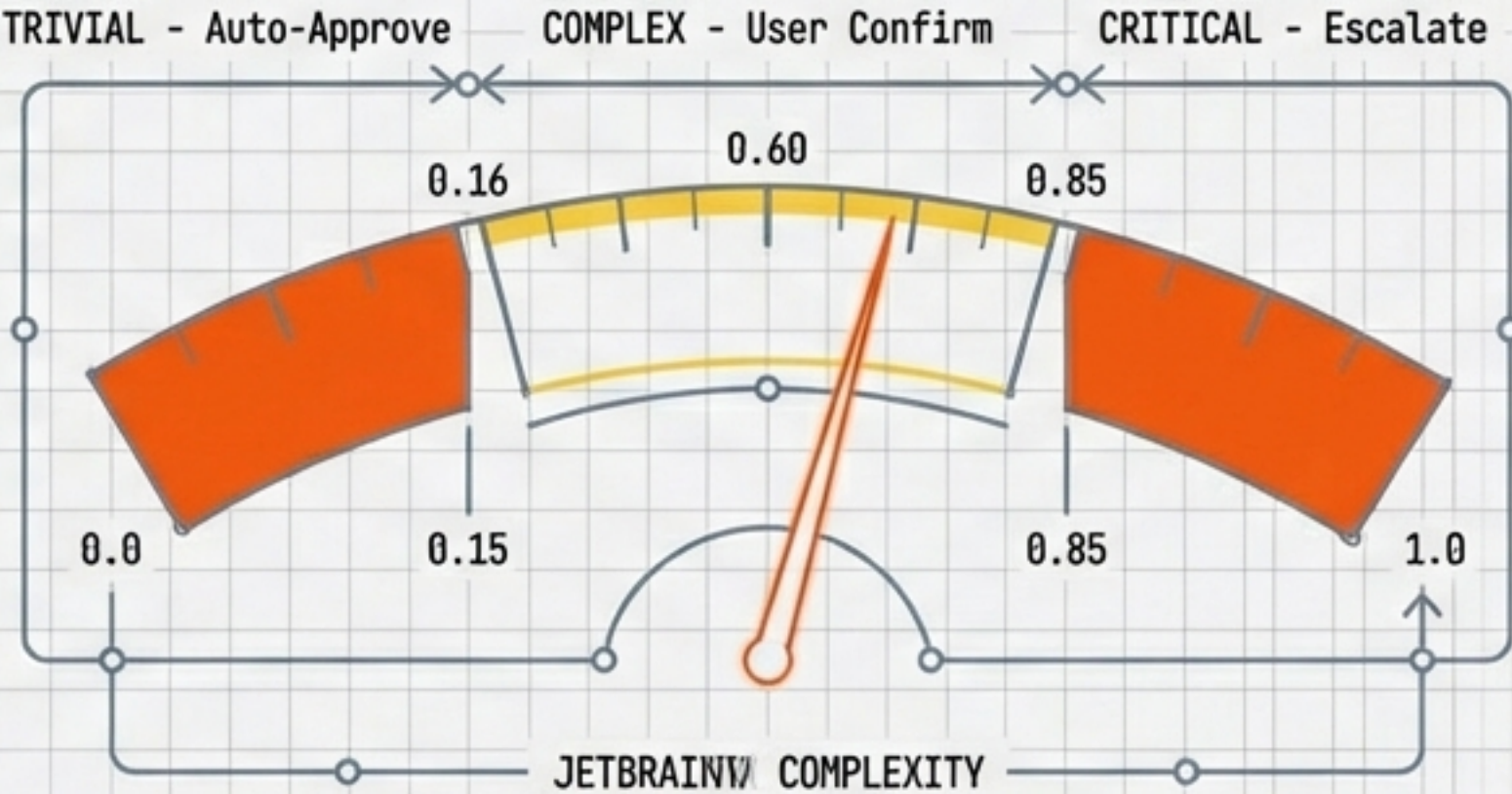
Intent Router Logic



Multi-Tier Governance Hierarchy



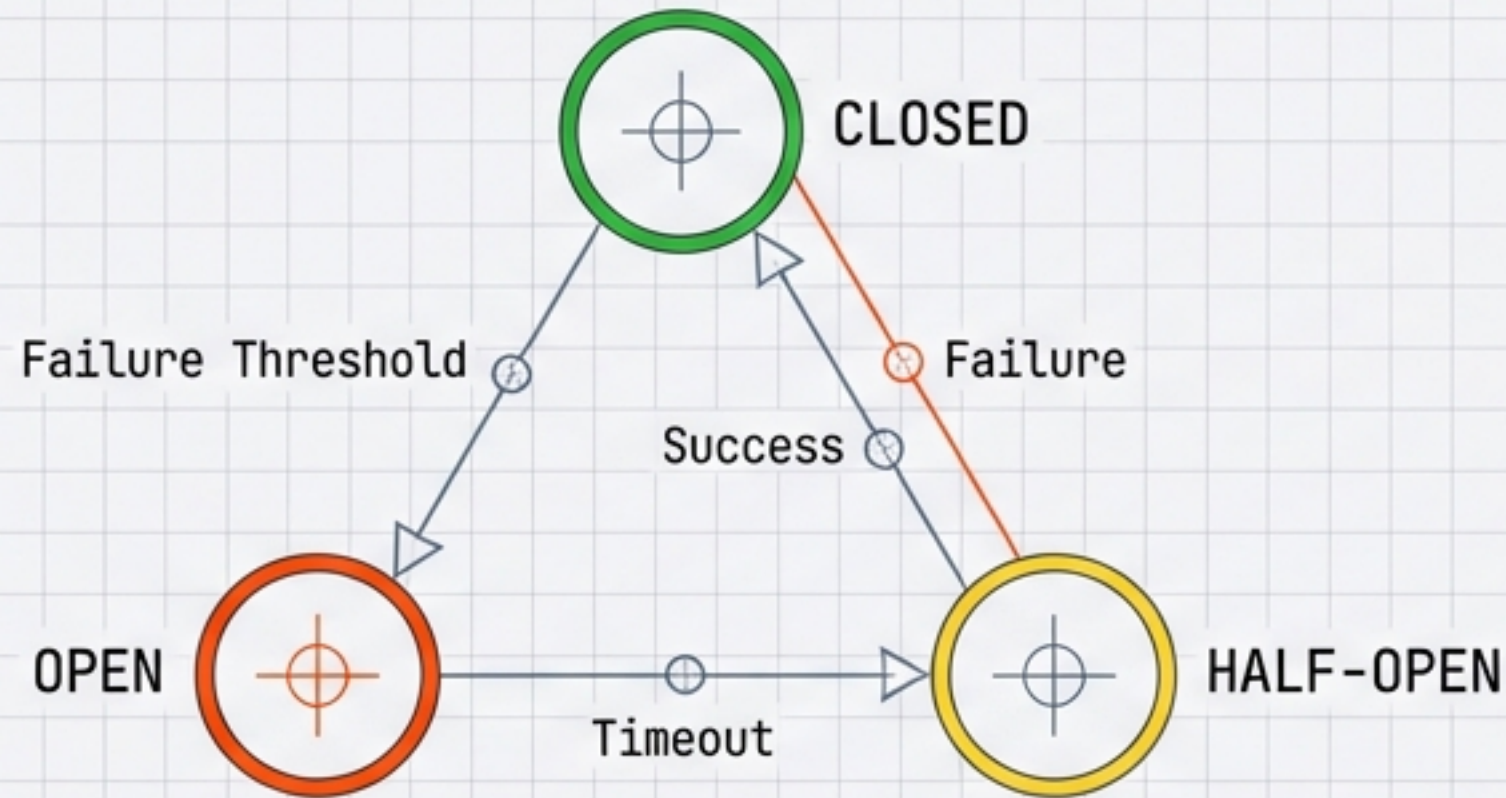
The Complexity Gate (Stage 2.5)



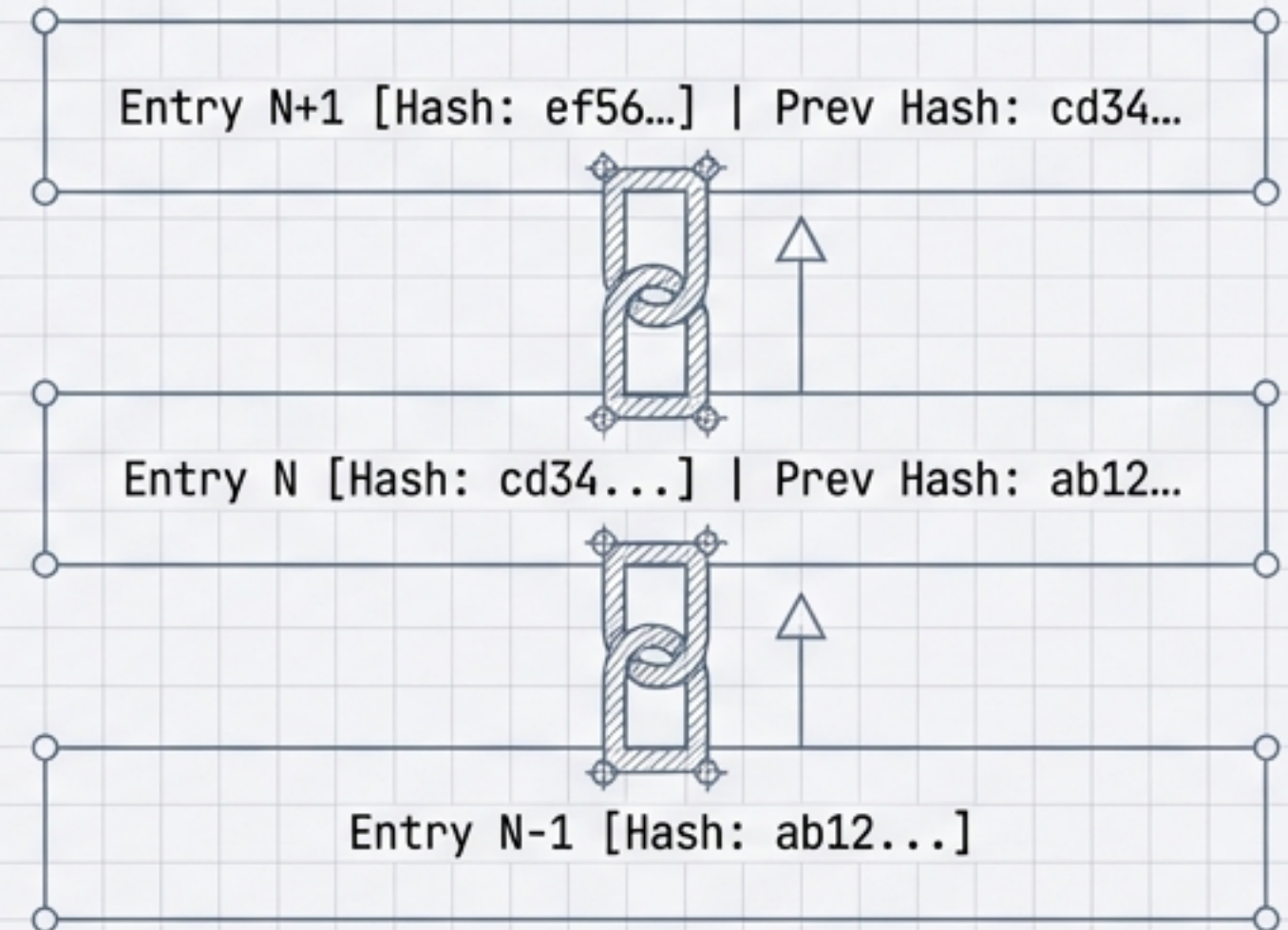
Resilience & Audit Integrity

Fail fast, degrade gracefully, recover automatically.

Circuit Breaker State Machine



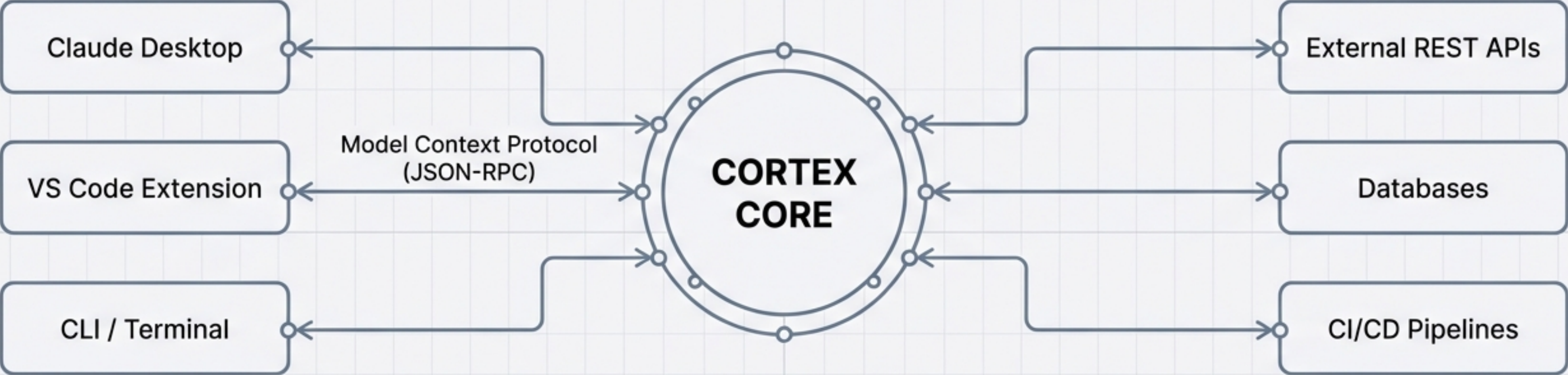
Audit Trail Architecture



Resilience Patterns

- ⊕ Partial Functionality: Degrade to cached knowledge
- ⊕ Bulkheads: Component isolation
- ⊕ Rollback: Atomic state recovery

Connectivity & MCP Ecosystem

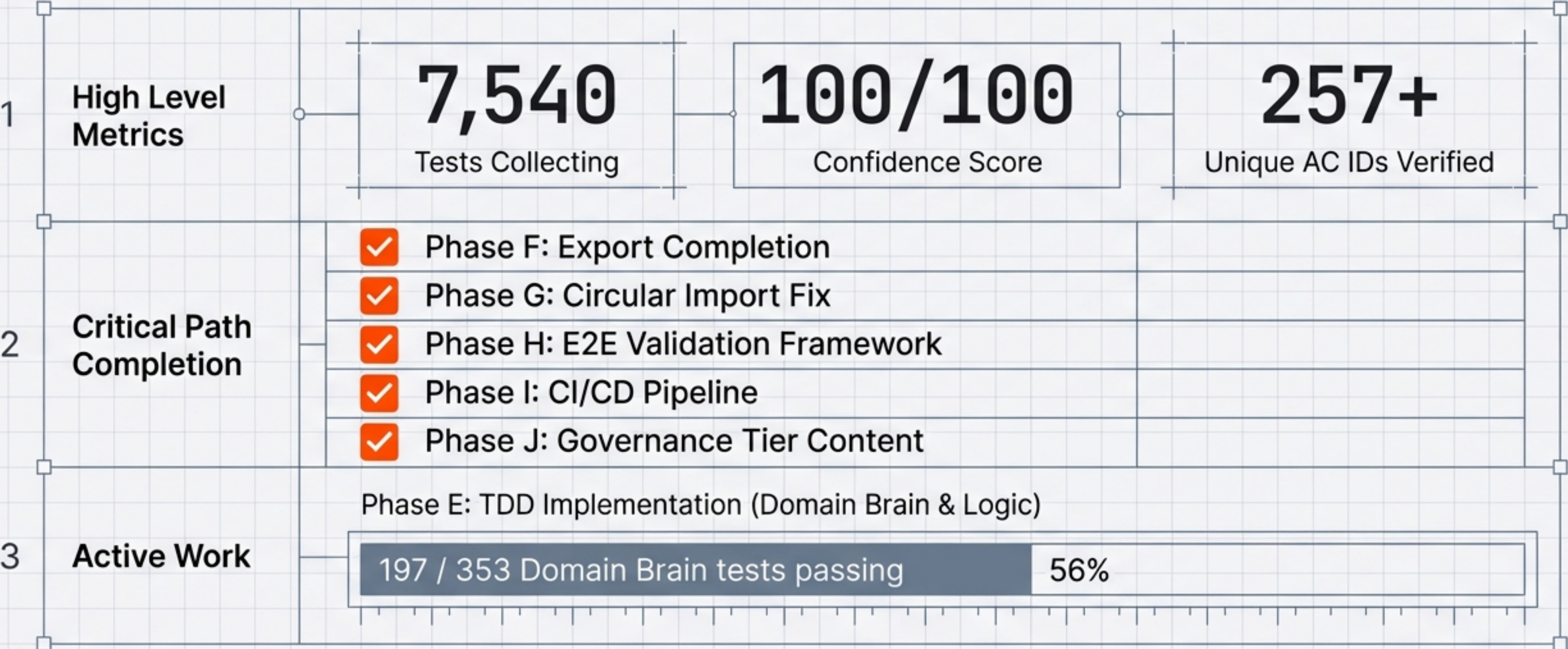


MCP Tool Suite (14 Registered Tools)

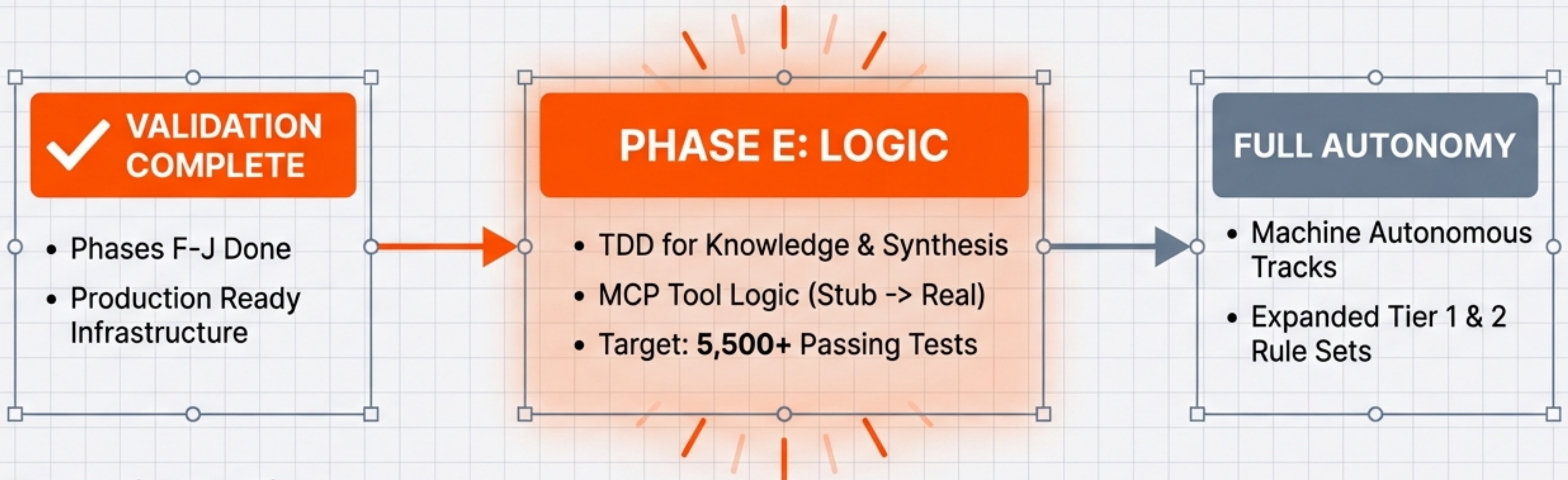
Governance	Orchestration	Knowledge
query-governance	monitor-orchestration	search-knowledge
validate-compliance	diagnose-issues	analyze-knowledge
execute-governance	status-orchestrator	generate-knowledge

Implementation Status

✓ **100% PRODUCTION READY (Phases F-J Complete)**



Roadmap to Full Autonomy



Immediate Actions

```
$ cortex health --verify
$ cortex audit list --limit 10
$ cortex orchestrator init MasterOrchestrator
```